

Student R&D-Project in Systems Engineering: Guidelines for Writing a Scientific or Technical Report.

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Abstract

This guideline is intended to provide a framework for manuscript writing in the context of student research projects (bachelor's thesis) in the systems engineering BSc programme. In the following, the essential parts (Introduction, Materials & Methods, Results, Discussion, and Conclusions) are described. The manuscript shall be logically structured along these parts: The introduction must include the research questions and purpose of the project. The Materials & Methods sections describe the research design and applied methods in a clear and reproducible manner. The results section presents the data without interpretations. The discussion is dedicated to a critical review of the applied methods and the validity of results. Finally, the conclusions shall be made on the basis of validated results, with regard to the limitations of the research design or applied methods.

The abstract is a summary of the project that should be structured in short paragraphs: Purpose(s), Materials & Methods, Results, and Conclusions. Keep the abstract brief and focus on the essential points of the project. Use neither figures, tables, lists, equations, nor references in the abstract. Describe the content in plain text only. A well-written abstract helps to convince the reader concerning the relevance and quality of the work.

Key Words: Bachelor's Thesis, Report Structure, Layout, Typography, AI Tools, Evaluation Criteria

Introduction

Scientific or technical report writing is a requirement in engineering. Therefore, writing a well-structured report according to scientific publication standards is a learning objective of the Systems Engineering BSc programme. This document is intended to give students advice on how to structure and write a well-understandable and clear report that matches scientific publication standards. These standards are not only valuable for scientific publications but also facilitate reading and understandability by focusing on a clear argumentation along the line: Purpose (what, which is the topic addressed, research questions, hypotheses) – Material & Methods (how it is done) – Results (what has been

observed) – Discussion (what are the limitations) – Conclusions. This scheme supports the argumentation along a clear logical line and should therefore be applied to every technical report as well. In the following, the different parts of a scientific or technical report are described in detail, including examples.

The section “Introduction” should include a brief description of the field of research or development. In this section, describe the state of research and the open issues that will be addressed by the presented project. The following points should be strictly considered:

- No replication of already published theory or findings; instead, provide references to these findings while describing the current state of research.
- Start with a more general introduction and then focus more and more on the research questions (like a funnel, starting with a more general view and then becoming more specific)
- End with clearly formulated statements concerning the research questions, purpose, or hypothesis

Avoid itemizations of bullet points: Use itemizations only if the clarity can be improved (e.g., as the aforementioned points). Describe the content in plain text, where every sentence is logically connected to the previous sentence (logical coherence of statements). This advice on avoiding bullet points also holds for the remaining parts of the report.

Citations and References

Every statement that is not derived by own, unpublished work (in a logical way) has to be referenced. References should include, whenever possible, original research papers or reviews. Textbooks are the second choice, and webpages should only be cited when a corresponding research paper is missing. Online publications should not be cited by a link to the webpage, where they are published, but by the DOI (s. examples in the References). References for product (manufacturer) information or information on a webpage may be included in footnotes, since these are not scientific papers. Scientific contents must be traceable. Scientific citation standards are intended to support this very important aspect of research, and therefore, the application of these standards is an essential learning objective of the Bachelor’s thesis.

There are several formats for citations. For the Bachelor’s thesis, students are free to select a standard, except if the supervisor insists on a certain standard. For scientific publications, the journals normally declare the required standards. Examples of citations are given in the following section, and the corresponding references can be found at the end of the manuscript.

When using citation styles with numbered in-text citations (e.g., [1]), the in-text citations should be given in ascending order, and the reference section should list the references in the same ascending order. When citing multiple sources simultaneously, the respective

citation numbers should be given in ascending order or as a range, i.e. ([1,2] or [1-3] instead of [1], [2] or [1],[2],[3]). In-text citations using author names and dates in parentheses, e.g., (Einstein, 1905), should be sorted in alphabetical order in the reference list.

In general, students are strongly encouraged to use software such as Zotero, Mendeley or EndNote to manage their references. Doing so will automatically maintain a consistent citation style and seamlessly renumber citations if needed. Using such software from the beginning of the project has the added benefit of allowing students to store a summary along with the reference, as well as a score on how relevant the document is to their project.

Example of an introduction

For learning about scientific publication standards, it is recommended to follow the structure and format of scientific papers, which may be used as sources in the own research project. The following section includes an example from Zellner et al. (2024):

“Computed tomography (CT) is the largest source of general and medical radiation exposure worldwide. The radiation dose delivered to patients in CT examinations is a public health concern [1,2]. Radiation protection principles like “as low as reasonably achievable” (ALARA) [3] and campaigns such as Image Gently [4] have led toward a better understanding of radiation exposure and have raised community awareness. Several studies have shown increased cancer risk in children who underwent CT and thus the importance of paediatric CT radiation protection [5,6]. The technical dose reduction methods are X-ray spectral beam shaping (e.g., tin filtering), detector efficiency improvements, and new image reconstruction methods, including iterative reconstruction algorithms and artificial intelligence methods [7,8]. Modern multidetector row CT devices allow helical and axial scanning. In wide detectors of up to 16 cm, axial scanning is possible without table movement and is therefore beneficial in paediatric imaging [9]. In recent literature, axial volumetric scan mode is described as the most dose-efficient scan mode [10]. Photon counting detector CT is rapidly emerging as a crucial modality in lung imaging, offering enhanced spatial resolution and potentially improved dose efficiency compared to conventional CT systems [11,12]. There is a wide variety of reported lung low-dose CT scan protocols in the current literature, with radiation doses ranging from minimal to significantly higher levels [11–14]. This study emphasizes the importance of establishing a threshold for providing ultra-low-dose lung CT scans that still yield diagnostic-quality images. To account for the variability in anatomical sizes, three differently sized phantoms were used in this study, allowing for a more comprehensive analysis of dose optimization across a range of body types.

The aim of the study was to find the lowest exposure dose providing diagnostic lung CT images using a deep learning-based image reconstruction algorithm. In addition, we

investigated and developed a quantitative method for assessing image quality of low-dose lung CT and defined dose thresholds necessary for achieving fully diagnostic lung images.”

Technical reports for product development projects

At first glance, product development projects seem to be different from research projects. However, a report structured along a logical line of argumentation is also a requirement for this type of project. The presented structure can also be applied to this type of project report. Typically, the introduction explains the background of the envisaged product, such as existing products on the market, existing prototypes, and their shortcomings, as well as the intended improvements.

The use of AI tools

AI tools can support both language/grammar as well as content generation. In case of language and grammar revisions, generative AI tools may improve readability and clarity, and their use can therefore be recommended to improve the manuscript. However, the proposed changes should be reviewed carefully since the meaning of a statement can change.

For content generation, AI tools should be used with caution. There are several shortcomings and problems:

First, the traceability of the content is limited, and even though the prompts allow the retrieval of references that match scientific criteria, the cited sources should always be verified and validated by reading the original paper. Generative AI can be very helpful for summarizing published papers, but reading the abstract may provide the same information. The added value of generative AI summaries can be seen in the possibility of asking for a summary regarding specific aspects.

Second, Large Language Models (LLMs) do not know the specific aspects of the bachelor's thesis project – otherwise, the project would not qualify as a research project. A well-written introduction summarizes the state of research regarding the aim and purpose of the project. Summaries generated by LLMs may be used as a starting point and should always be regarded as a proposition that needs intellectual digestion and adaptation to one's own ideas. Otherwise, there is a risk of useless replication of already known things (for the community, for the author, these contents may be unknown, but a scientific or technical report addresses messages to the community).

Third, text generated with LLMs tends to exhibit outstanding wordsmithing, but unless meticulous care is taken, the resulting paragraphs will lack a guiding thread and lead to a poorly written report in consequence.

Lastly, LLMs typically generate language that sounds plausible, but may be humbug upon closer inspection. Students are held accountable for their work. Thus, the report must not contain such wording, and students must comprehend the content of their report and must be able to explain their work in their own words.

Materials & Methods

The Material & Method chapter is dedicated to a reproducible description of how the own experiments, measurements, developments, etc. are implemented and performed. This may include the theoretical foundation of the research or development methods used in the own project. However, the explanations of well-known or previously established (published) theory should be limited to a minimum required to understand the way, how the research or the development process was conducted. Do not anticipate any results in this chapter.

In the Materials & Methods chapter, the following aspects may be included:

- methodological approaches (research design)
- description of used models (geometry, physics, numerical implementation)
- derivation of own models (physical, mathematical, and numerical aspects)
- specific methods for product development
- description of experimental setups or measurement setups
- methods of verification and validation (includes models as well as products such as hardware, software, or other deliverables)
- applied numerical methods
- applied statistical methods
- In case of software projects, an overview of the implementation (code may be presented in an appendix)
- in case of reviews and literature studies, the search methods and pipeline, keywords and prompts used for search, etc.

Data sheets of technical components and used sensors, or material lists, should be included in the Appendix. The methods shall be described in plain text in a clear, reproducible, and logical way. Illustrative pictures, diagrams, or mathematical equations can be supportive for the understanding. Pictures should be included only in the manuscript when they provide added value.

Avoid excessive numbering of chapters, sections, and paragraphs (e.g. 2.12.7a Method A) and select meaningful titles and headings (e.g. "Derivation of Viscosity Model" instead of "Theoretical Calculation"). The readability can be improved by a first paragraph or section explaining the structure of the work (research design, structure of workflows, or data flows, the embedding of different parts such as modelling, implementation, simulations, verification, validation, or measurements). This part can be followed by sections dedicated to the specific aspects of the applied methods. In general, it does not make sense to have

a single subsection within a section. In that case, the subsection title can be omitted after naming the section appropriately.

The layout and typography are intended to increase readability. The column width should be adapted to the font size. A ratio of column width to font size larger than 1.5 cm/points should be avoided (in this document, the ratio of column width to font size is 1.27 cm/points, and 1.5 for the figure captions, respectively).

Equations shall be structured clearly. Equations are not simply wild clusters of symbols, but have a mathematical or physical meaning. By grouping the variables and constants, the readability can be improved. As an example, the coulomb force $\vec{F}_E$ between two electrical charges q_1 and q_2 in a vacuum is given by:

$$\vec{F}_E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 \cdot q_2}{r_{12}^2} \cdot \frac{\vec{r}_{12}}{|\vec{r}_{12}|} \quad (1)$$

In Eq.1, ϵ_0 is the vacuum permittivity (electrical constant), and $\vec{r}_{12} = \vec{r}_1 - \vec{r}_2$ is the difference of the position vector of q_1 and q_2 . By numbering the equations, they can be referenced in the text by Eq.XY. Equation numbers should be set in ascending order and assigned automatically by Word, Latex etc. Doing so avoids tedious and error-prone manual renumbering attempts in cases where equations are added and removed during the report writing. Do not directly insert values or numbers, but use clearly defined variables. It is important to use a consistent notation. Students should, e.g., use either f or ν for frequency throughout the report instead of using one notation in a section and the other notation in other sections. This holds in particular when working in groups. A table with the chosen notation can be added to the front matter of the thesis to further enhance clarity. The variables shall be described in plain text, explaining the definitions or models behind them, if needed. Definitions of variables in tables should be an exception and may be used in cases of models having a large number of variables.

Tab.1. Variables/parameters and constants used in Eq.1 and the range of values used for the measurements and model fitting in Fig.1.

Variable / unit	Description	Value (range)
ϵ_0 / F·m ⁻¹	<i>Vacuum permittivity</i>	$8.854 \cdot 10^{-12}$
q_1 / C	Electrical charge 1	$1.0 \cdot 10^{-10}$ - $2.0 \cdot 10^{-10}$
q_2 / C	Electrical charge 2	$1.0 \cdot 10^{-10}$ - $2.0 \cdot 10^{-10}$
r_{12} / m	Distance between charges	$1.0 \cdot 10^{-6}$ - $1.0 \cdot 10^{-5}$

Every table must be numbered and must have a table header above the table (see Tab.1 as an example). Every table must be referenced, for example, as (Tab.X), at least once in the text. To distinguish table headers and figure captions, use a different font size. Always leave sufficient space between tables, figures, and the manuscript text. Make sure to provide units where needed and to round numbers appropriately.

Results (& Discussion)

In the results section, the results (observations) shall be presented objectively without any interpretation. This may include data from experiments such as measurements or output of computer simulations, or, in the case of product development projects, verification and validation data of final designs. To reach a logical and clear presentation of the results, an aggregation of the measurements or simulation output may be helpful. For example, for extensive computer simulations, periodic signals in the time domain may be transformed to the frequency domain, and these frequencies may be displayed as a function of varied parameters. For data presentation, select an appropriate form (table, diagrams, heat maps, etc.). Carefully generated, illustrative figures are very supportive in presenting results. It is recommended not to start with writing the results section until good figures, described by clear figure captions, have been elaborated.

Figures/diagrams shall support the message (observed behaviour, dependence of a simulated or observed variable on another variable, etc.) in a clear and illustrative manner. As an example, Fig.1 exhibits measurements and model fits for the electrical force according to Eq.1 and Tab.1. In analogy to tables, a reference (Fig.X or Figure X) must be placed in the text at least once. Text and figures shall correspond in content, but since figures have to be numbered and referenced by the number, figures can be placed independently of the corresponding paragraph in the manuscript text. However, figures should not be placed far away from the corresponding content (as illustrated by Fig.1 in this document).

Every figure must be described by a figure caption, starting with the figure number. A well-crafted figure caption has a title, followed by a brief description of what can be seen in the figure. If a figure consists of multiple subfigures, they can be addressed individually after an initial, general description. Every figure axis must be labeled with the parameter/variable (divided by the corresponding unit since the axis typically shows only the dimensionless value) displayed on the corresponding axis. Ensure that axis labels and numbers are set in a sufficiently large font to maintain readability. For measured points, the uncertainty (standard errors, standard deviations) shall be displayed by error bars. For this, measurements should be taken at least in triplicate (five or more measurements per point allow a more or less robust estimation of the standard deviation).

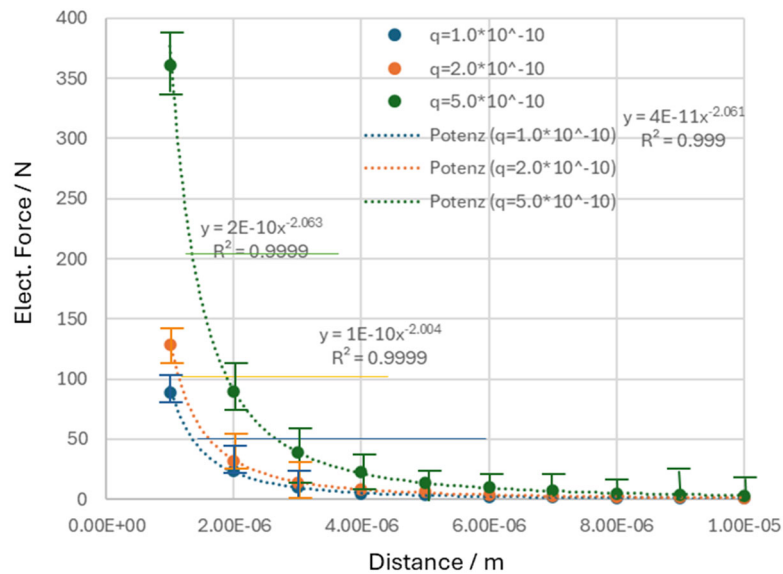


Fig.1. Measurements and model fitting of the electrical force between two electrical charges: The points indicate the measured values (error bars correspond to the standard error); the polynomial fits are in reasonably agreement with Coulomb's law.

Instead of connecting measured points by straight lines, models can be fitted to the experimental points. This increases not only the clarity of plots (e.g., which data points belong to each other) but also provides a deeper insight into possible dependencies. The easiest way may be the use of data-descriptive models such as polynomial, exponential, power law, or logarithmic functions. If the physical, chemical, or biological background of the investigated system is known, a dynamic model may be fitted to the data for checking the agreement with theory.

Fig.1 exhibits some flaws: First, it contains a mix of German ("Potenz") and English (axis labels) language, which should be avoided. Since such plots have to be redrawn often, at times shortly before submission of the report, students should store the scripts to generate their data along with the data to be able to implement changes effortlessly. Second, the graphs cannot be distinguished well at larger distances. An option for displaying graphs with largely varying values is the use of logarithmic scales. For exponential functions, plots with a logarithmic y-axis result in a straight line. For power laws, a double logarithmic plot also converts the graphs into straight lines (Fig.2). The slope is defined by the exponent and can be determined graphically. In addition, the small values are represented clearly, and deviations from the theoretical model (the Coulomb law in Fig.2) become visible by varying slopes of the straight lines and misfits of points.

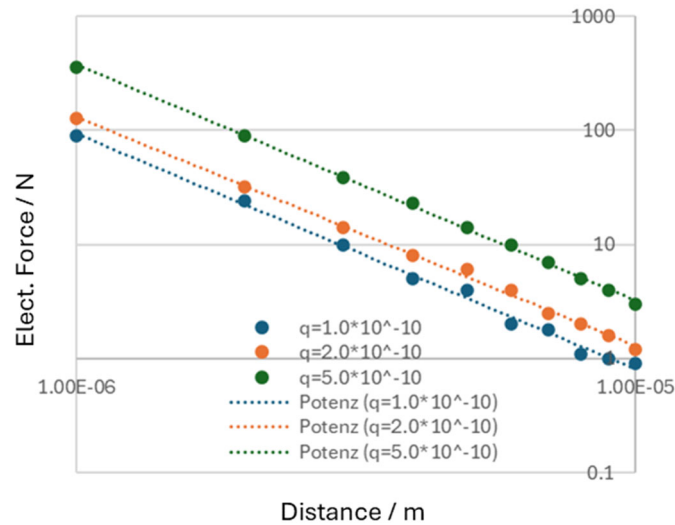


Fig.2. Measurements and model fitting of the electrical force between two electrical charges, as displayed in Fig.1: The three lines corresponding to electrical charges of $1.0 \cdot 10^{-10}$, $2.0 \cdot 10^{-10}$, and $5.0 \cdot 10^{-10}$ C; are clearly separated, the deviations from the straight lines and slightly varying slopes of the power-law fits in the double-logarithmic plot indicate small differences from Coulomb's law.

(Discussion &) Conclusions

Normally, discussion and conclusions are grouped in one chapter. Sometimes, a discussion closely related to the presented results may be beneficial, especially in the case of projects with several extensive subprojects. However, it is important to clearly present results objectively and free of interpretations. A main point of the discussion is a critical view of the reliability and validity of the presented work. This includes a critical review of the used models and methods, the agreement with theory or the work of other authors, or general weaknesses of experiments, measurements, numerical approaches, etc.

Conclusions shall be concise and, if possible, quantitative. Concluding statements have to be limited to the range of validity. Make these statements based on validated data or results. Assumption and guesswork may be part of the discussion and have to be clearly visible as such. Give answers to the research questions and confirm the hypotheses stated in the introduction. The conclusions shall address every research question and vice versa. During a research project, new questions may appear. This is common in a typical research project since research is always like a journey into unknown terrain. Therefore, the research questions or hypotheses in the introduction may be refined or adapted, as long as the research design is adequate. It is important to note that a scientific report does not reproduce the chronological order of the project (work packages) but presents a conclusive description of what (Introduction) was, how (Materials and Methods) done and observed (Results), and what can be concluded based on the project results (Conclusions).

In the Appendix, the criteria for evaluation of a student research project are provided. These points not only include criteria for the evaluation of the report, but also points that can be applied to the course of the project.

References

Reference Style 1

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Appendix

Criteria for evaluation of a student's research or development project:

(A) Independent scientific thinking/originality: Does the thesis contain significant independent contributions by the student? Does the thesis reflect some originality / new ideas by the student?

(B) General scientific and methodological competence: Has the student acquired sufficient knowledge and familiarity with the literature? Does the student state the goals of the thesis and the questions to be addressed clearly (Refined research questions have to be based on the problem statement given above)? Are the results discussed in a broader context and in view of previous work? Are the methods and techniques used properly described? Are the methods used appropriate for the purpose of the thesis? Have the methods/results been properly analysed and tested for their validity? Are the limitations of the results analysed and clearly stated? Has the research been carried out carefully? Can the results be trusted? Does the student give an outlook on possible subsequent research? Did the student learn or even develop new methods or tools (design of experiments, measurement techniques, program codes, etc.)?

(C) Logical coherence / Quality of presentation: Does the thesis include the mandatory elements: abstract, introduction, materials & methods, results, and conclusion? Is the logical structure of the thesis appropriate for the subject? Does the thesis reflect previous work sufficiently and distinguish its own contribution clearly from already known studies? Are facts clearly distinguished from assumptions and hypotheses? Is the work presented in a way that enables reproducibility of the results? Are the results clearly discussed in the context of the questions asked? Does the conclusion provide more insights than a simple list of the results (key points learned from the study)? Is the text correct language-wise, comprehensible, and scientifically sound?

(D) Work process: Did the student show dedication towards the research task? Did the student acquire the necessary competence to pursue his project? Did the student evolve positively in terms of independence and maturity? Did the student make good use of criticism and instructions? Did the student require intensive guidance?